

## Sheet 4

$$1) S \rightarrow 0S1 \mid \epsilon$$

Derivation

$$\begin{aligned} S &\rightarrow 0S1 \\ &\rightarrow 00S11 \\ &\rightarrow 000S111 \\ &\rightarrow 00001111 \end{aligned}$$

$$\begin{aligned} 2) A &\rightarrow 0A0 \\ &\rightarrow 01A10 \\ &\rightarrow 01110 \end{aligned}$$

$$\begin{aligned} 3) S &\rightarrow (S) \\ &\rightarrow (SS) \\ &\rightarrow ((S)(S)) \\ &\rightarrow ((( ))) \end{aligned}$$

$$\begin{aligned} 4) S &\rightarrow 0S1 \\ &\rightarrow 00S11 \\ &\rightarrow 000S111 \\ &\rightarrow 000A111 \\ &\rightarrow 0000A111 \\ &\rightarrow 00000111 \end{aligned}$$

belong to the language

$$5- S \rightarrow aSb \mid A$$

$$A \rightarrow aA \mid \epsilon$$

$$6- S \rightarrow \langle a \rangle S \langle /a \rangle \mid \langle P \rangle S \langle /P \rangle$$

$$\mid \text{text} \mid \epsilon$$

7) right

$$\begin{aligned} S &\rightarrow aSS \\ &\rightarrow aS\epsilon \\ &\rightarrow aaAS \\ &\rightarrow aaAaSS \\ &\rightarrow aaAaSE \\ &\rightarrow aaAa\epsilon \\ &\rightarrow aa\bar{b}aa \end{aligned}$$

left

$$\begin{aligned} S &\rightarrow aSS \\ &\rightarrow aaAS \\ &\rightarrow aabaSS \\ &\rightarrow aaba\bar{a}SSS \\ &\rightarrow aabaa\epsilon SSS \\ &\rightarrow aabaa\epsilon\epsilon S \\ &\rightarrow aabaa\epsilon\epsilon\epsilon \end{aligned}$$

$$8) a) S \rightarrow A1A1A1A$$

$$A \rightarrow 0A \mid 1A \mid \epsilon$$

$$b) (0/1)^i 0 (0/1)^i ; i \geq 0$$

$$S \rightarrow 0S1 \mid 0S0 \mid 1S1 \mid 1S0 \mid \epsilon$$

$$\begin{aligned}
 9) \quad S &\rightarrow AX \mid TC \\
 A &\rightarrow aA \mid \epsilon \\
 X &\rightarrow bXC \mid \epsilon \\
 T &\rightarrow aTb \mid \epsilon \\
 C &\rightarrow cC \mid \epsilon
 \end{aligned}$$

it is Ambiguous; because there are 2 different ways  
 to produce any word  $a^x b^x c^x$  ie number of  
 $a$ 's =  $b$ 's =  $c$ 's such as  $aaa bbb ccc$

$$\begin{aligned}
 10) (a) \quad S &\rightarrow Aa \\
 A &\rightarrow abA \mid \epsilon
 \end{aligned}$$

$$\begin{aligned}
 b) \quad S &\rightarrow aA \\
 A &\rightarrow abA \mid ab
 \end{aligned}$$

$$\begin{aligned}
 11) \quad S &\rightarrow AB \\
 &\rightarrow \overbrace{aAA} B \\
 &\rightarrow a \underbrace{aA} a \downarrow b
 \end{aligned}$$

$$\begin{aligned}
 S &\rightarrow AB \\
 &\rightarrow \overbrace{aA} B \\
 &\rightarrow a \overbrace{aA} B \\
 &\rightarrow aa \underbrace{a} \downarrow b
 \end{aligned}$$

12) ① Since the start appeared in the R.H.S; then  
Create new start

$$T \rightarrow A \quad A \rightarrow BAB \mid B \mid \epsilon$$

2) eliminate null  
روح لک production فی A و شیل

①  $A \rightarrow \epsilon$

$$T \rightarrow A \mid \epsilon$$

$$A \rightarrow BAB \mid B \mid B \mid B \mid \epsilon$$

②  $B \rightarrow \epsilon$

روح لک production فی B و شیل

$$A \rightarrow BAB \mid AB \mid BA \mid B \mid BB$$

$$B \rightarrow \epsilon$$

new Grammar

$$T \rightarrow A \mid \epsilon$$

$$B \rightarrow \epsilon$$

$$A \rightarrow BAB \mid BA \mid AB \mid B \mid BB$$

3) ای حاجہ فی اکثرین non terminal  
A → ABAB

$$\Rightarrow X \rightarrow BA \quad \& \quad A \rightarrow XB$$

ای حاجہ فی اکثرین terminal  
production

$$B \rightarrow \epsilon \Rightarrow B \rightarrow YY$$

$$Y \rightarrow \epsilon$$

## new Grammar

$$T \rightarrow A \mid \epsilon \quad \text{Production في أول E لا حذف}$$

$$A \rightarrow X \cancel{A} B \mid A B \mid B A \mid B \mid B B$$

$$B \rightarrow Y Y \quad Y \rightarrow \emptyset$$

$$X \rightarrow B A$$

### 4) Remove units

يعني أي Production فيه non terminal واحد  
نحذفه ونحذف ما يقابله

$$A \rightarrow X B \mid A B \mid B A \mid B B \mid Y Y$$

$$T \rightarrow X B \mid A B \mid B A \mid B B \mid Y Y \mid \epsilon$$

$$B \rightarrow Y Y \quad Y \rightarrow \emptyset \quad X \rightarrow B A$$

$$13) (1) S \rightarrow S \quad \leadsto \text{Creat new start}$$

$$(2) S \rightarrow A S A \mid a B \Rightarrow$$

$$X \rightarrow a$$

### 3) Remove $\epsilon$

$$S \rightarrow A S A \mid X B \mid X$$

$$B \rightarrow b$$

$$A \rightarrow B \mid S \mid \epsilon$$

$\Rightarrow$  Remove new  $\epsilon$  in A

remove  $\epsilon$  in A

$$S \rightarrow ASA \mid SA \mid AS \mid \boxed{S} \mid X \mid XB$$

$$A \rightarrow B \mid S \quad B \rightarrow b$$

$$S' \rightarrow S \quad X \rightarrow a$$

split ASA  $\Rightarrow Y \rightarrow AS$

$$S' \rightarrow S \quad A \rightarrow B \mid S$$

$$B \rightarrow b \quad X \rightarrow a$$

$$S \rightarrow AS \mid SA \mid A \mid X \mid XB$$

remove unit (unary)

$$A \rightarrow b \mid AS \mid SA \mid YA \mid a \mid XB$$

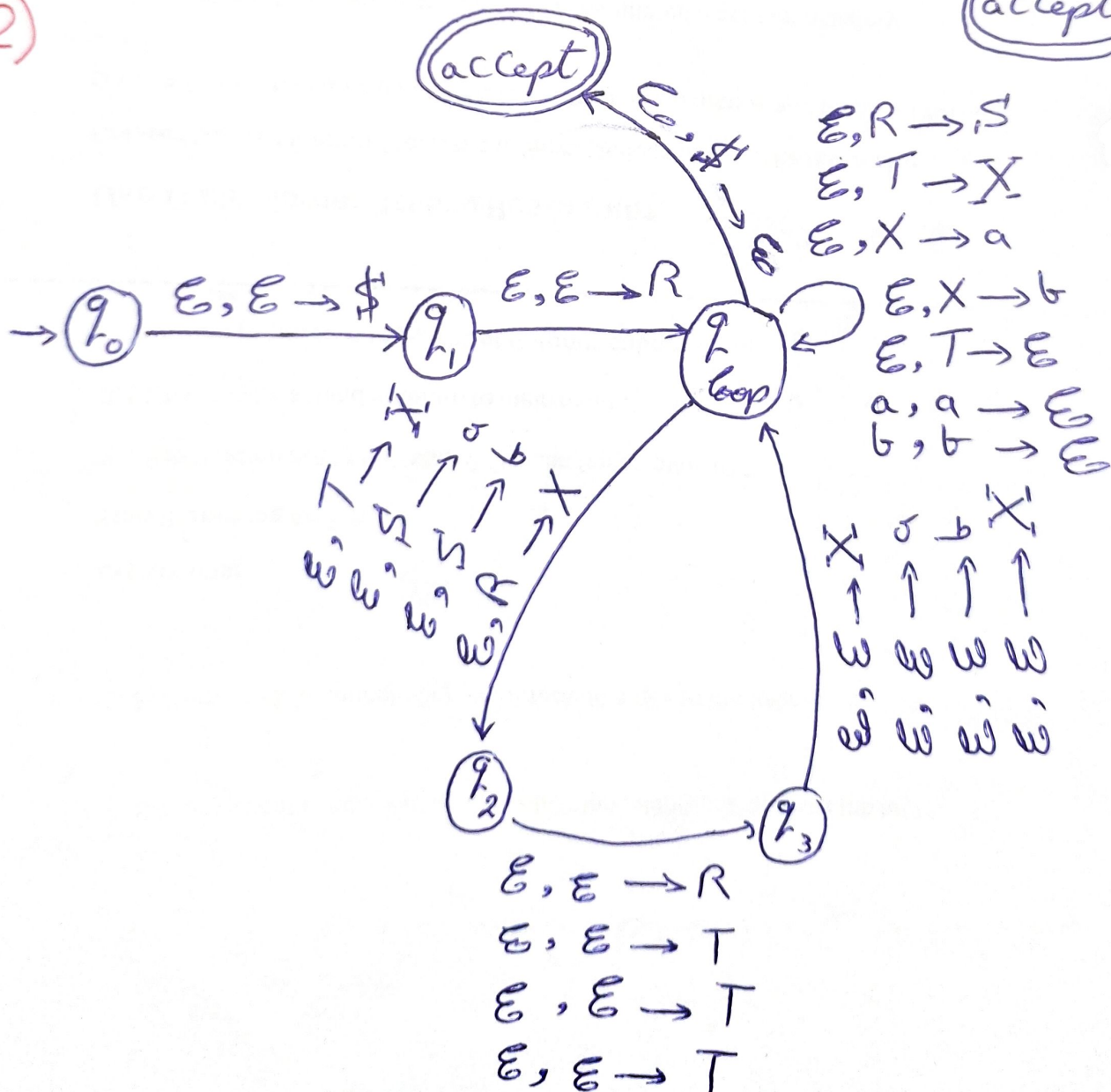
$$S \rightarrow AS \mid SA \mid YA \mid a \mid XB$$

$$S' \rightarrow AS \mid SA \mid YA \mid a \mid XB$$

$$B \rightarrow b \quad X \rightarrow a$$

$$Y \rightarrow AS$$

1)



3)

$$\delta(q_1, 221, \epsilon) = \{(q_1, S\#)\}$$

$$\delta(q_1, 221, S) = \{(q_1, SB\#)\}$$

$$\delta(q_1, 221, B) = \{(q_1, BBB\#)\}$$

$$\delta(q_1, 221, B) = \{(q_1, 2BBS\#)\}$$

$$\delta(q_1, 21, 2) = \{(q_1, BBS\#)\}$$

$$\delta(q_1, 21, B) = \{(q_1, 2BS\#)\}$$

$$\delta(q_1, 1, 2) = \{(q_1, BS\#)\}$$

$$\delta(q_1, \epsilon, 1) = \{(q_1, S\#)\}$$

$$\delta(q_1, \epsilon, S) = \{(q_1, A\#)\}$$

$$\delta(q_1, \epsilon, A) = \{(q_1, \epsilon\#)\}$$

$$\delta(q_1, \epsilon, \#) = \{(q_2, \epsilon, \epsilon)\}$$

accept

# S

read  $\epsilon$ , Pop  $\epsilon \rightarrow$  Push  $S$

# S B

$\epsilon$ ,  $S \rightarrow SB$

# S 1 B B

$\epsilon$ ,  $B \rightarrow 1 B B$

# S 1 B 2

$\epsilon$ ,  $B \rightarrow 2$

# S 1 B

read 2, Pop 2  $\rightarrow$  push  $\epsilon$

# S 1 2

# S 1

# S

# A

# ε

Read ε, Pop B → Push 2

2, 2 → ε

1, 1 → ε

ε, S → A

ε, A → ε

ε # ε

accept